Linear Regression Analysis of Story Features

Introduction

This analysis investigates whether specific narrative features can predict the overall story score of posts and comments from the ChangeMyView dataset. The dataset contains a variable called story_score, which measures the strength of storytelling on a scale from 0 to 1. Additionally, six narrative features are included: agency_score, event_score, world_score, curiosity_score, surprise_score, and suspense_score, each measured on a scale from 1 to 5. The goal of this analysis is to determine whether any of these story features can act as a predictor for the story score using simple linear regression.

Dependent and Independent Variables

In this analysis, the dependent variable is story_score, as it represents the value we aim to predict. The independent variables are the individual narrative features: agency_score, event_score, world_score, curiosity_score, surprise_score, and suspense_score. Each of these variables is tested separately to determine whether it can predict the story score.

Scatterplot Analysis

Scatterplots were generated using Matplotlib to visually examine the relationship between each independent variable and the story score. In these plots, the independent variable is shown on the x-axis and the story score on the y-axis. Scatterplots help identify whether there is a visible linear relationship between the variables.

Most of the scatterplots show a more or less dispersed distribution of data points, suggesting that the relationship between the narrative features and the story score may be weak. Some variables appear to show slight positive trends, meaning that higher feature scores might correspond to slightly higher story scores. However, the relationship does not appear strong based solely on visual inspection.

Linear Regression

Simple linear regression models were fitted for each independent variable separately. The slope indicates how much the story score changes when the feature score increases by one unit. A positive slope indicates that higher feature scores are associated with higher story scores. The intercept represents the predicted story score when the feature value is zero. While the intercept is part of the regression equation, it is often less meaningful for interpretation in this context because the feature scores typically range between 1 and 5.

Regression lines were plotted together with the scatterplots to visualize how well the linear model fits the data.

R^2 Score

The R^2 score was calculated for each regression model using the sklearn.metrics function. The R^2 value measures how well the independent variable explains the variation in the dependent variable. An R^2 value closer to 1 indicates a strong predictive relationship, while a value closer to 0 indicates a weak relationship.

The R^2 values obtained in this analysis were relatively low, suggesting that none of the individual narrative features strongly predict the story score on their own. This indicates that story quality may depend on a combination of narrative features rather than a single factor.

Residual Analysis

Residual plots were generated to evaluate the differences between the observed story scores and the predicted values from the regression models. Residuals represent the errors in the model's predictions.

In the residual plots, the residuals appear scattered around zero without forming a strong pattern. This suggests that the linear model is reasonably appropriate for exploring the relationship between the variables, although the predictive strength remains limited.

Shapiro-Wilk Test

The Shapiro-Wilk test was performed on the residuals of each regression model to determine whether the residuals follow a normal distribution. The test produces a p-value that indicates whether the assumption of normality is satisfied. If the p-value is greater than 0.05, the residuals are considered approximately normally distributed.

In every case, the p-values were above 0.05, indicating that the residuals follow a normal distribution. This suggests that the normality assumption of linear regression is satisfied for these models.

Comparison of the Individual Variables

The results of the regression analyses were compared across the different storytelling features based on the scatterplots, regression lines, R^2 values, and residual analyses produced for each variable.

The scatterplots show that most variables have a fairly scattered distribution of points, indicating only weak linear relationships with the story score. The regression lines also show relatively small slopes, suggesting that changes in the individual storytelling features lead to only small changes in the predicted story score.

The R^2 scores further confirm this observation. The values remain relatively low across the different variables, indicating that none of the storytelling features alone guarantees an increase in the story score. The residual plots also show a wide spread of residuals around zero, which suggests that the predictions of the models are limited in accuracy.

Conclusion

Overall, the analysis suggests that none of the individual narrative features strongly predict the story score when examined independently. Although some variables may show slight positive relationships with the story score, the low R^2 values indicate that a single feature alone cannot explain much of the variation in storytelling strength. It is likely that the story score is influenced by a combination of multiple narrative elements, as well as other factors that are not included in the dataset.